Stochastic process simulation and Euler-Maruyama scheme

Jan Rems

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1 Deep BSDE

Exercise 1.1. *The CIR model describes the evolution of the interest rate. It is given by the SDE*

$$dX_t = a(b - X_t)dt + \sigma\sqrt{X_t}dB_t, \qquad X_0 = x_0,$$

for $a, b, \sigma > 0$*. Given* Y_t *a zero-coupon bond paying 1 at maturity* T*, we have*

$$Y_t = \mathbb{E}\Big[\exp\Big(-\int_t^T X_s ds\Big)\Big|\mathcal{F}_t\Big].$$

Actually, Y_t *satisfies the BSDE*

$$Y_t = 1 - \int_t^T X_s Y_s ds - \int_t^T Z_s dBs.$$

Assuming $a = b = \sigma = T = x_0 = 1$ *estimate* Y_0*. The true value is* $Y_0 \approx 0.3965$*. Draw graphs of loss and* Y_0 *values over 3000 epochs. Use batch size 1000, 100 time points, and the hidden dimension 11. All the processes are 1 dimensional.*

Exercise 1.2. *We deal with the example of a linear BSDE*

$$-dY(t) = \theta Z(t)dt - Z(t)dB(t), \quad Y(T) = X,$$

where we consider the terminal condition $X = \exp(\theta B(T) - \frac{\theta^2}{2}T)$ *and* $\theta \in \mathbb{R}$*. Using formula for linear BSDEs and the martingale property of* X*, compute the explicit solution and compare it to the numerical one. Use* $T = 1$ *and* $\theta = 0.2$*.*

2 Deep Stochastic Control

Exercise 2.1. *Let* $dX_t = X_t u_t[bdt + \sigma dB_t]$*,* $X_0 = x_0$ *be a controlled wealth process with portfolio* u*. Construct a deep learning algorithm that estimates this control given value function*

$$V(x) = \sup_u E[\log(X_T)],$$

for $T = x_0 = 1$, $b = 0.05$, $\sigma = 0.2$.

Exercise 2.2. *Let* $dX_t = X_t[(b - c(t))dt + \sigma dB_t]$, $X_0 = x_0$ *be a controlled consumption process with consumption* c. *Construct a deep learning algorithm that estimates the consumption given value function*

$$V(x) = \sup_c E\Big[\int_0^T \log(c(t)X_t)dt + \log(X_T)\Big],$$

for $T = x_0 = 1$, $b = 0.05$, $\sigma = 0.2$.